Programming Assignment B: Longest EVEN–ODD Substring

Problem Description

Write a Java program that finds, for each input string, the longest contiguous substring belonging to the language EVEN–ODD. EVEN–ODD consists of all strings over {a, b} having an even number of 'a' and an odd number of 'b'. If several substrings attain the maximum length, choose the rightmost one (the one with the largest start index).

Program Interface

Command line: `java LongestEvenOdd <inputFile> <outputFile>`

- The program will be invoked with exactly two arguments: input filename and output filename. You do not need to check the number of arguments; if incorrect, it may crash.
- Input file format:
  - Line 1: a single integer N (N ≥ 0)
  - Lines 2..N+1: each line contains a single string (may contain characters other than 'a' or 'b').
- Output file format: N lines of output. For each input string, in order:
  - If the string contains an EVEN–ODD substring, write a single line containing `<substring> <startIndex> <endIndex> <length>` for the longest one (rightmost on ties).
  - Otherwise, write a single line containing `NONE`.

Examples

Example input file (input.txt):

```
8
ab
aaa
abba
bbbb
ababab
bxaabaa
abaxaba
abbbab
```

Example output file (output.txt):

```
b 1 2 1
NONE
b 2 3 1
bbb 1 4 3
babab 1 6 5
aabaa 2 7 5
aba 4 7 3
abbba 0 5 5
```

Requirements

- You must use `java.util.regex.Pattern` and `java.util.regex.Matcher`.
- Your program must identify EVEN–ODD substrings by matching against a regular expression, not by counting characters. Tallying the number of 'a's and 'b's, or tracking their parity, to decide which substrings belong to EVEN–ODD — rather than applying the pattern — is not allowed, even though it would produce the same answers.
- Indices are 0-based, and `endIndex` is exclusive (the Java `substring` convention).
- Characters other than 'a' or 'b' may appear in the input. They cannot appear inside an EVEN–ODD substring and may separate matches.